

Zackary Rackauckas

GENAI RESEARCH SCIENTIST (NLP SPEECH)

Corona del Mar, California, USA

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Education

Columbia Engineering

MS COMPUTER SCIENCE, THESIS TRACK

New York, NY

December 2025

- Thesis advisor: Prof. Julia Hirschberg.
- GPA: 4.0/4.0
- Research areas: Computer Assisted Language Learning, Speech, Dialogue Systems in Education, Large Language Models.

The University of Tokyo

SPECIAL GRADUATE RESEARCH STUDENT

Tokyo, Japan

May 2025 - July 2025

- Supervisor: Nobuaki Minematsu.
- Research areas: Computer Assisted Language Learning, Human-computer Interaction, Human-agent Interaction.

Swarthmore College

BA MATHEMATICS

Swarthmore, PA

May 2023

- Minors in Computer Science and Japanese.
- GPA: 3.8/4.0.
- Research areas: Natural Language Processing, Sentiment Analysis, Japanese Linguistics.

Professional Experience

The University of Tokyo

VISITING RESEARCH INTERN

Tokyo, Japan

May 2025 - July 2025

- Conducted research in the Minematsu Lab on multimodal language learning systems, focusing on integrating conversational AI into STEM education environments to enhance English-medium instruction.

RoleGaku

FOUNDER

New York, NY

Dec. 2023 - Present

- Leading teams of mobile developers, voice actors, artists, server administrators, writers, and web developers to create Jouzu, a Japanese learning app centered around LLM-driven voiced character chatbots.
- Creating 30 scripted lessons and LLM-based algorithms for in-text word inspection.
- Developing and deploying backend REST APIs leveraging LLMs, speech synthesis models, speech recognition models, SQL databases, and cutting-edge NLP techniques using Python, Flask, GitHub, and cloud hosting.

Infineon Technologies

RESEARCH INTERN

San Jose, CA

Aug. 2023 - Apr. 2024

- Developed a RAG-based product chatbot to provide comprehensive product information, reducing engineer product lookup time from several minutes to seconds.
- Leveraged machine learning systems to make predictions based on user web activity to optimize product targeting strategies, now implemented for 6 major customers.
- Collaborated with Infineon's generative AI team on natural language processing competitor analysis.

Stanford University School of Medicine

INTERN

Stanford, CA

Jul. 2021 - Dec. 2021

- Developed an interactive website to publish with the Tool for Evaluation of Media Portrayals of Suicide.
- Let journalists verify whether their article on suicide was helpful or harmful and gave instructions to rewrite their article before publication.

Research Experience

Columbia University - Dept. of Computer Science

New York, NY

SUPERVISOR: DR. JULIA HIRSCHBERG

Aug. 2024 - Present

- Project: Created AdaptLingo, a chatbot for English learning that adapts to users' capabilities through prosody detection and constrained decoding of LLMs. Developed a pronunciation detection and correction algorithm using Flow Matching TTS algorithms and speech editing for English learners.

Columbia University - Dept. of Computer Science

New York, NY

SUPERVISOR: DR. JULIA HIRSCHBERG

Aug. 2024 - Present

- Thesis: Voiced Character Language Chatbots for Japanese Learning through an LLM- and Speech Processing-Driven Mobile App.

University of Tokyo - Dept. of Electrical Engineering

Tokyo, Japan

SUPERVISOR: DR. NOBUAKI MINEMATSU

May 2025 - July 2025

- Project: Created and published Re:Member, a QA dialogue system that leverages videos of personal memories for Japanese learning. Mentored a Bachelor's student to create LangInLab, a system for augmenting engineering lab Instruction with vision- and voice-enabled agents in preparation for the University of Tokyo's shift to English as a Medium of Instruction.

Chapman University - Machine Learning and Affiliated Technologies Lab

Orange, CA

SUPERVISOR: DR. ERIK Linstead

Jun. 2022 - Aug. 2022

- Project: Analyzed sentiment to discover changes in public opinions on Twitter before and after the overturning of Roe v Wade, and wrote a paper targeting submission to the Journal of Big Data.

Swarthmore College - Dept. of Modern Languages and Literature

Swarthmore, PA

SUPERVISOR: DR. JOHN BUNDSCHUH

Jun. 2022 - Aug. 2022

- Project: Digitized old Japanese Buddhist sutras to research change in the Japanese language, especially verb forms, from archaic to modern language.

Chapman University - Dept. of Mathematics

Orange, CA

SUPERVISOR: DR. DANIELE STRUPPA

Mar. 2018 - Aug. 2018

- Project: Transcribed bicomplex number research documents in LaTeX and assisted with collaborative analysis problems.

Publications**JOURNAL ARTICLES (PEER-REVIEWED)**

Zackary Rackauckas. 2025. Learner Interactions with Jouzu, the Mobile Application with Conversational Characters. *JALT-CALL Trends*. URL: <https://doi.org/10.29140/jct.v1n1.2164>

Zackary Rackauckas. 2024. RAG-Fusion: A New Take on Traditional RAG. *International Journal on Natural Language Computing*, 13(1), 11. URL: <https://doi.org/10.5121/ijnlc.2024.13103>

CONFERENCE AND WORKSHOP PAPERS (PEER-REVIEWED)

Zackary Rackauckas, Julia Hirschberg. 2025. Comparative Evaluation of Expressive Japanese Character Text-to-Speech with VITS and Style-BERT-VITS2. Accepted to *IEEE Annual Ubiquitous Computing, Electronics and Mobile Communication Conference (UEMCON 2025)*. URL: <https://arxiv.org/abs/2505.17320>

Zackary Rackauckas, Nobuaki Minematsu, Julia Hirschberg. 2025. Re:Member: Emotional Question Generation from Personal Memories. Accepted to *4th Workshop on Bridging Human-Computer Interaction and Natural Language Processing (HCI+NLP) at the Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*. URL: <https://aclanthology.org/2025.hcinlp-1.13.pdf>

Zackary Rackauckas, Julia Hirschberg. 2025. VoxRAG: A Step Toward Transcription-Free RAG Systems in Spoken Question Answering. In *Proceedings of the 1st Workshop on Multimodal Augmented Generation via Multimodal Retrieval (MAGMAR 2025) at the Annual Meeting of the Association for Computational Linguistics (ACL 2025)*. URL: <https://aclanthology.org/2025.magmar-1.3.pdf>

Zackary Rackauckas, Arthur Camara, Jakub Zavrel. 2024. Evaluating RAG-Fusion with RAGElo: An Automated Elo-Based Framework. In *Proceedings of the 1st Workshop on Large Language Models for Evaluation in Information Retrieval (LLM4Eval) at the ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2024)*. URL: <https://ceur-ws.org/Vol-3752/paper6.pdf>

EXTENDED ABSTRACTS (PEER-REVIEWED)

Masanori Shigi, **Zackary Rackauckas**, Yuka Akiyama, Nobuaki Minematsu. 2025. LangInLab: Augmenting Engineering Lab Instruction with Vision- and Voice-Enabled AI Agents for Language Learning. Accepted to *International Conference on Human-Agent Interaction (HAI 2025)*.

PREPRINTS AND UNDER REVIEW

Zackary Rackauckas, Julia Hirschberg. 2025. Learning Japanese with Jouzu: Interaction Outcomes with Stylized Dialogue Fictional Agents.. URL: <https://arxiv.org/abs/2507.06483>

Zackary Rackauckas, Daniel Lee, Siyan Li, Julia Hirschberg. 2025. AdaptLingo: An Adaptive Speech-to-Speech Chatbot to Match Users' English Proficiency. (In Progress) .

Presentations

Zackary Rackauckas, Julia Hirschberg. 2025. Comparative Evaluation of Expressive Japanese Character Text-to-Speech with VITS and Style-BERT-VITS2. *IEEE 16th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference*, New York, USA. **Awarded Best Presentation**

Zackary Rackauckas, Julia Hirschberg. 2025. "VoxRAG: A Step Toward Transcription-Free RAG Systems in Spoken Question Answering" Poster Presentation: *1st Workshop on Multimodal Augmented Generation via Multimodal Retrieval (MAGMaR 2025)*, Vienna, Austria.

Zackary Rackauckas, 2025. "Demonstrating an AI Chatbot that Adapts to Users' English Proficiency with End-to-end Speech" Workshop presentation: *Japan Association for Language Teaching – Computer-Assisted Language Learning (JALTCALL 2025)*, Tokyo, Japan.

Zackary Rackauckas, 2025. "Information Retrieval and Dialogue Systems for Computer-assisted Language Learning" Oral presentation: *GAVO Minematsu & Saito Laboratory*, Tokyo, Japan.

Zackary Rackauckas, Riya Raj, Siyan Li, Julia Hirschberg. 2025. "AdaptLingo: An Adaptive Speech-to-Speech Conversational Agent Tailored to Users' English Proficiency Levels." Poster presentation: *Columbia Data Science Day*, New York, USA.

Zackary Rackauckas 2025. "Evaluating Jouzu: A Fun and Engaging Japanese Learning App" Oral presentation: *Advanced Topics in Spoken Language Processing*, New York, USA.

Zackary Rackauckas, Riya Raj, Siyan Li, Julia Hirschberg. 2025. "AdaptLingo: An Adaptive Speech-to-Speech Conversational Agent Tailored to Users' English Proficiency Levels." Poster presentation: *Columbia AI Summit*, New York, USA.

Zackary Rackauckas. 2024. "Jouzu: Voiced Character Language Chatbots for Japanese Learning." Oral presentation: *Computer-Enabled Abilities Laboratory*, New York, USA.

Zackary Rackauckas, Arthur Camara, Jakub Zavrel. 2024. "Evaluating RAG-Fusion with RAGElo: an Automated Elo-based Framework." Poster and oral presentations: *The First Workshop on Large Language Models (LLMs) for Evaluation in Information Retrieval (LLM4Eval @ SIGIR)*, Washington, D.C., USA.

Zackary Rackauckas. 2024. "Voiced Character Language Chatbots for Japanese Learning," Oral presentation: *Japan Association for Language Teaching – Computer-Assisted Language Learning (JALTCALL 2024)*, Nagoya, Japan.

Awards & Certifications

2025	Best Presentation Award , IEEE UEMCON	
2025	Friends of Global UTokyo Leadership Award , The University of Tokyo	\$ 4,000
2024	Japanese Language Proficiency Test, N2 , The Japan Foundation	
2024	Artificial Intelligence (AI) for Business , Wharton Online	
2021 - 2023	Centennial Academic Honor Roll , Centennial Conference	

Professional Affiliations and Review Activities

- 2025 **The Thirteenth International Conference on Human-Agent Interaction**, Reviewer
- 2025 **Demonstrating an AI Chatbot that Adapts to Users' English Proficiency with End-to-end Speech**, Workshop organizer and presenter
- 2025 **The Third Workshop on Large Language Models (LLMs) for Evaluation in Information Retrieval**, Program committee, reviewer
- 2025 **The Second Workshop on Large Language Models (LLMs) for Evaluation in Information Retrieval**, Program committee, reviewer
- 2025 **JALTCALL Trends**, Reviewer for the journal's inaugural edition
- 2025 - Now **ACL Student Member**, Kerrville, TX
- 2024 - Now **IEEE Student Member**, Santa Clara Valley Section
- 2023 - Now **IEEE Computer Society Member**, San Jose, CA
- 2023 - 2024 **IEEE Professional Member**, Santa Clara Valley Section

Skills

TECHNICAL

- Natural Language Processing, Speech Processing, Machine Learning, Computer Assisted Language Learning, Chatbots.

PROGRAMMING LANGUAGES

- Python, C/C++, R, Java, HTML/CSS/JS, Perl.

FRAMEWORKS AND TOOLS

- Pytorch, Scikit-learn, Tensorflow, NLTK, Flask, Praat.

LANGUAGES

- English, Japanese, Mandarin Chinese, Spanish.

Mentoring

- 2024 **Shigi Masanori**, BS Student, The University of Tokyo
- 2024-2025 **Riya Raj**, MS Student, Columbia University